

# Spring 1981

**Problem 1 (Sp81).** Let  $\vec{i}$ ,  $\vec{j}$ , and  $\vec{k}$  be the usual unit vectors in  $\mathbb{R}^3$ . Let  $\vec{F}$  denote the vector field

$$(x^2 + y - 4)\vec{i} + 3xy\vec{j} + (2xz + z^2)\vec{k}$$

1. Compute  $\nabla \times \vec{F}$  (the curl of  $\vec{F}$ ).
2. Compute the integral of  $\nabla \times \vec{F}$  over the surface  $x^2 + y^2 + z^2 = 16$ ,  $z \geq 0$ .

**Problem 2 (Sp81, Sp22).** Let  $T$  be a linear transformation of a vector space  $V$  into itself. Suppose  $x \in V$  is such that  $T^m x = 0$ ,  $T^{m-1}x \neq 0$  for some positive integer  $m$ . Show that  $x, Tx, \dots, T^{m-1}x$  are linearly independent.

**Problem 3 (Sp81).** Let  $D$  be an ordered integral domain and  $a \in D$ . Prove that  $a^2 - a + 1 > 0$ .

**Problem 4 (Sp81).** Consider the system of differential equations

$$\begin{aligned}\frac{dx}{dt} &= y + x(1 - x^2 - y^2) \\ \frac{dy}{dt} &= -x + y(1 - x^2 - y^2)\end{aligned}$$

1. Show that for any  $x_0$  and  $y_0$ , there is a unique solution  $(x(t), y(t))$  defined for all  $t \in \mathbb{R}$  such that  $x(0) = x_0$ ,  $y(0) = y_0$ .
2. Show that if  $x_0 \neq 0$  and  $y_0 \neq 0$ , the solution referred to in Part 1 approaches the circle  $x^2 + y^2 = 1$  as  $t \rightarrow \infty$ .

**Problem 5 (Sp81).** Decompose  $x^4 - 4$  and  $x^3 - 2$  into irreducible polynomials over  $\mathbb{R}$ , over  $\mathbb{Z}$ , and over  $\mathbb{Z}_3$ .

**Problem 6 (Su77, Sp81).** Let  $\hat{a}_0 + \hat{a}_1 z + \dots + \hat{a}_n z^n$  be a polynomial having  $\hat{z}$  as a simple zero. Show that there is a continuous function  $r : U \rightarrow \mathbb{C}$ , where  $U$  is a neighborhood of  $(\hat{a}_0, \dots, \hat{a}_n)$  in  $\mathbb{C}^{n+1}$ , such that  $r(a_0, \dots, a_n)$  is always a zero of  $a_0 + a_1 z + \dots + a_n z^n$ , and  $r(\hat{a}_0, \dots, \hat{a}_n) = \hat{z}$ .

**Problem 7 (Sp81, Sp10, Sp19, Sp22).** Evaluate  $\int_{-\infty}^{\infty} \frac{x \sin x}{(1 + x^2)^2} dx$ .

**Problem 8 (Sp81).** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous and  $k \in \mathbb{N}$ . Prove that there is a real polynomial  $P(x)$  of degree  $\leq k$  which minimizes (for all such polynomials)

$$\sup_{0 \leq x \leq 1} |f(x) - P(x)|$$

**Problem 9 (Sp81).** Show that the following three conditions are all equivalent for a real  $3 \times 3$  symmetric matrix  $A$ , whose eigenvalues are  $\lambda_1, \lambda_2$ , and  $\lambda_3$ :

1.  $\text{tr } A$  is not an eigenvalue of  $A$ .
2.  $(a + b)(b + c)(a + c) \neq 0$ .
3. The map  $L : S \rightarrow S$  is an isomorphism, where  $S$  is the space of  $3 \times 3$  real skew-symmetric matrices and  $L(W) = AW + WA$ .

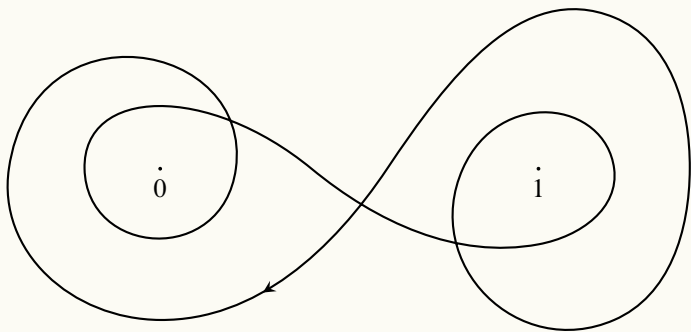
**Problem 10 (Sp81, Sp24).** 1. Give an example of a sequence of  $\mathcal{C}^1$  functions

$$f_k : [0, \infty) \rightarrow \mathbb{R} \quad k = 1, 2, 3, \dots$$

such that  $f_k(0) = 0$  for all  $k$ , and  $f'_k(x) \rightarrow f'(x)$  for all  $x$  as  $k \rightarrow \infty$ , but  $f_k(x)$  does not converge to  $f(x)$  for all  $x$  as  $k \rightarrow \infty$ .

2. State an extra condition which would imply that  $f_k(x) \rightarrow f(x)$  for all  $x$  as  $k \rightarrow \infty$ .

**Problem 11 (Sp81).** Evaluate  $\int_C \frac{e^z - 1}{z^2(z - 1)} dz$  where  $C$  is the closed curve shown below:



**Problem 12 (Sp81).** For  $x \in \mathbb{R}$ , let

$$A_x = \begin{pmatrix} x & 1 & 1 & 1 \\ 1 & x & 1 & 1 \\ 1 & 1 & x & 1 \\ 1 & 1 & 1 & x \end{pmatrix}$$

1. Prove that  $\det(A_x) = (x-1)^3(x+3)$ .
2. Prove that if  $x \neq 1, -3$ , then  $A_x^{-1} = -(x-1)^{-1}(x+3)^{-1}A_{-x-2}$ .

**Problem 13 (Sp81, Sp22).** Which of the following series converge?

$$1. \sum_{n=1}^{\infty} \frac{(2n)!(3n)!}{n!(4n)!} \quad 2. \sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}} \quad 3. \sum_{n=2}^{\infty} \frac{1}{n^{1+1/\log^2 n}}$$

**Problem 14 (Sp81).** The set of real  $3 \times 3$  symmetric matrices is a real, finite dimensional vector space isomorphic to  $\mathbb{R}^6$ . Show that the subset of such matrices of signature  $(2, 1)$  is an open connected subspace in the usual topology on  $\mathbb{R}^6$ .

**Problem 15 (Sp81, Fa11).** Let  $\mathbb{K}$  be one of the fields  $\mathbb{C}$ ,  $\mathbb{R}$ ,  $\mathbb{Q}$ ,  $\mathbb{F}_{4044121}$  (the finite field with  $4044121 = 2011^2$  elements). For which of the above choices of  $\mathbb{K}$  is the ring  $\mathbb{K}[x]/\langle x^4 + 6x - 12 \rangle$  a field?

**Problem 16 (Sp81).** Let  $f(x)$  be a real valued function defined for all  $x \geq 1$ , satisfying  $f(1) = 1$  and

$$f'(x) = \frac{1}{x^2 + f(x)^2}$$

Prove that  $\lim_{x \rightarrow \infty} f(x)$  exists and is less than  $1 + \frac{\pi}{4}$ .

**Problem 17 (Sp81).** Let  $b$  be a real nonzero  $n \times 1$  matrix. Set  $M = bb^\top$ .

1. Prove that there is an orthogonal matrix  $Q$  such that  $QMQ^{-1} = D$  is diagonal, and find  $D$ .
2. Describe geometrically the linear transformation  $M : \mathbb{R}^n \rightarrow \mathbb{R}^n$ .

**Problem 18 (Sp81).** Describe the two regions in  $(a, b)$ -space for which the function

$$f_{a,b}(x, y) = ay^2 + bx$$

restricted to the circle  $x^2 + y^2 = 1$ , has exactly two, and exactly four critical points, respectively.

**Problem 19 (Sp81).** Let  $G$  be a finite group. A *conjugacy class* is a set of the form

$$C(a) = \{bab^{-1} \mid b \in G\}$$

for some  $a \in G$ .

1. Prove that the number of elements in a conjugacy class divides the order of  $G$ .
2. Do all conjugacy classes have the same number of elements?

**Problem 20 (Sp81).** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous with  $f(0) = 0$ . Show there is a continuous concave function  $g : [0, 1] \rightarrow \mathbb{R}$  such that  $g(0) = 0$  and  $g(x) \geq f(x)$  for all  $x \in [0, 1]$ .

A function  $g : I \rightarrow \mathbb{R}$  is *concave* if

$$g(tx + (1-t)y) \geq tg(x) + (1-t)g(y)$$

for all  $x$  and  $y$  in  $I$  and  $0 \leq t \leq 1$ .